Lecture 5 v2:
Multivariate Linear Regression

25117 - Econometrics

Universitat Pompeu Fabra

October 16th, 2024

What we learned in the last lesson

- Hypothesis testing for regression coefficients is analogous to hypothesis testing for the population mean: Use the t-statistic to calculate the p-values and either accept or reject the null hypothesis. Like a confidence interval for the population mean, a 95% confidence interval for a regression coefficient is computed as the estimator ± 1.96 standard errors.

- When X is binary, the regression model can be used to estimate and test hypotheses about the difference between the population means of the "X $= 0$" group and the "X $= 1$" group.

- Estimate the coefficient covariance matrix with $SER^2(\mathbf{X}'\mathbf{X})^{-1}$.

What we learned in the last lesson

- Under the maintained OLS assumptions, as a result of the Gauss–Markov theorem, the OLS estimator is BLUE.

- If the maintained OLS assumptions hold and the regression errors are normally distributed, then the OLS t-statistic computed using the usual OLS standard errors has a Student t distribution when the null hypothesis is true. The difference between the Student t distribution and the normal distribution is negligible if the sample size is moderate or large.

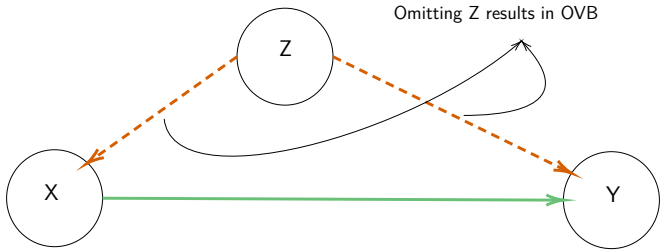
Omitted Variable Bias (OVB)

- If β_1 is a causal effect, the first assumption of the Gauss-Markov Theorem must hold : $\mathbb{E}(u_i \mid X_i) = 0$ for all i
- Remember that $\hat{\beta}_1 = \beta_1 + \frac{cov(X,u)}{var(X)}$ and $\hat{\beta}_1$ will be unbiased only if $\mathbb{E}(\frac{cov(X,u)}{var(X)}) = 0$
- **Example:**
  - Private university graduates tend to have higher wages than public university graduates. Does that mean that private university education (X_1) yields higher wages (Y)?
  - Maybe, but not necessarily — Private university graduates are *selected* by those institutions
  - They differ in a number of ways from public university graduates: they have on average better test scores in highschool, they come from wealthier families, etc.
  - All these relevant but unobserved variables may be omitted, and, if not accounted for, may bias the estimation of β_1
  - In comparing wages from private and public university graduates, how to *identify* the causal effect of education (X_1) and differentiate it from the effect of, say, advantageous social background?
  - We want to make apple-to-apple comparisons...

Omitted Variable Bias (OVB)

The bias in the OLS estimator that occurs as a result of an omitted factor, or variable, is called omitted variable bias. For omitted variable bias to occur, the omitted variable Z must satisfy two conditions:

1. Z is a determinant of Y (i.e. Z is part of u); and
2. Z is correlated with the regressor X (i.e., $corr(Z, X) \neq 0$)

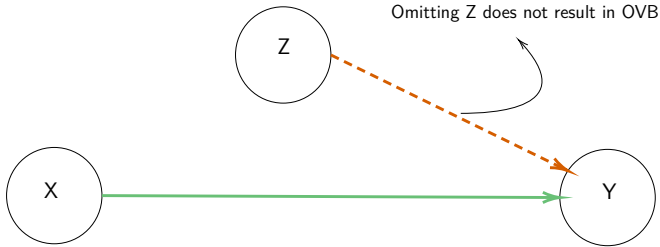


Both conditions must hold for the omission of Z to result in omitted variable bias.

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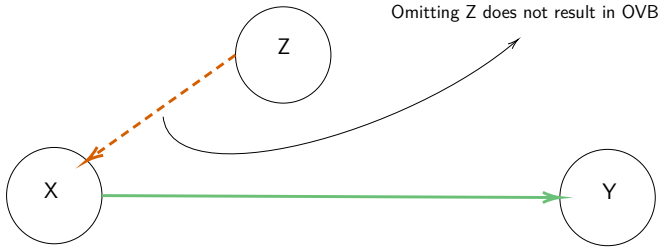


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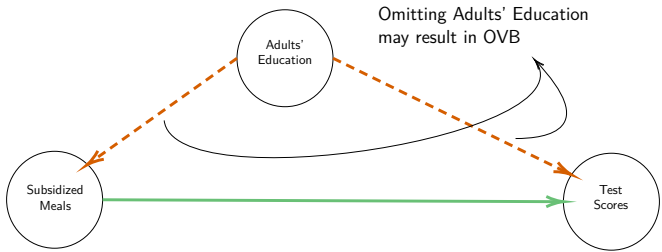


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Omitted Variable Bias (OVB)

In the California Schools example,

1. Adults' educational attainment may affect local kids' test scores; and
2. Adults' educational attainment is a driving force of local income, and, therefore, likely affects the share of subsidized meals



Again, both conditions must hold for the omission of Z to result in omitted variable bias.

Omitted Variable Bias (OVB)

There are systematic differences in educational attainment across districts with low and high share of subsidized meals.

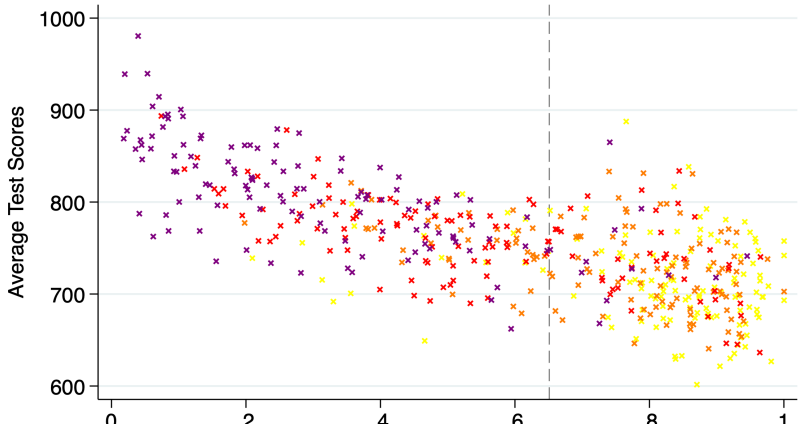
- Districts more educated adults have higher test scores
- Districts less educated adults have more schools with subsidized meals
- Among districts with comparable adults' educational attainment, the effect of subsidized meals is generally smaller

	Above-median % Subsidized meals		Below-median % Subsidized meals		Difference in Test Scores
	$\overline{Y}$	n	$\overline{Y}$	n	$\overline{Y}^1 - \overline{Y}^0$
All Districts	720.2472	250	787.4612	250	67.214
By Share w/ College Education:					
[0; .1339436]	710.3194	108	745.3444	18	35.025
[.1350751; .1996677]	724.0533	90	758.4917	36	34.43834
[.1997542; .3260502]	731.9475	40	774.4559	84	42.50845
[.3309678 ; .7696404]	742.05	12	813.2955	112	71.24553

Another way to see this

The scatterplot shows one point per school:

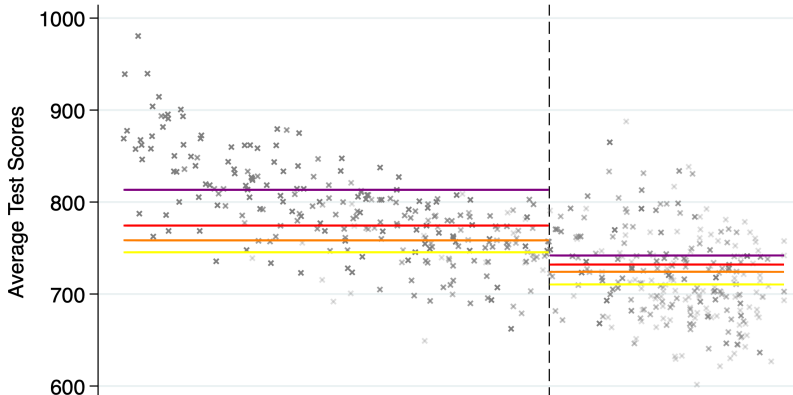
- Horizontal axis: share of subsidized meals (X); vertical axis: average test score (Y).
- Each color represents a quartile of the share of adults with a college education.
- Comparing all points mixes schools with different education levels, so the raw relationship can combine the meal effect with differences in background characteristics.



Another way to see this

The colored horizontal lines are group-specific average test scores:

- Within each education quartile, the line summarizes the average score for schools with a given level of subsidized meals.
- The vertical dashed line separates the above-median and below-median subsidy groups.
- The lines show that education level shifts test scores, so comparing the two subsidy groups without controlling for education is not an apples-to-apples comparison.



What to do about it?

- Randomization implies that any differences between the treatment and control groups are random — not systematically related to the treatment
- From now on, one of the main goals of this course is to explore ways to recover causal treatment effects
- Remember, we want to make ceteris paribus, apple-to-apple comparisons — not pear-to-apple, etc.
- One way to do it is to **control** for systematic differences between control and treatment group
- In our example, we want to know the average treatment effect of subsidizing school meals on test scores, holding the share of college education, average income, English proficiency, etc. constant (i.e., getting closer to all else equal, or ceteris paribus)
- Said differently, we want to know the average impact of a change in subsidized meals on test scores within groups where college education, average income, English proficiency, etc. is the same

Example for $k = 2$

Imagine we compare two schools, i and j

$$\begin{cases} Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} \\ Y_j = \beta_0 + \beta_1 X_{1j} + \beta_2 X_{2j} \end{cases}$$

$$\Rightarrow \begin{cases} Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} \\ Y_i + \Delta Y_i = \beta_0 + \beta_1 (X_{1i} + \Delta X_{1i}) + \beta_2 (X_{2j} + \Delta X_{2i}) \end{cases}$$

$$\Rightarrow \Delta Y_i = \beta_1 \Delta X_{1i} + \beta_2 \Delta X_{2i}$$

When X_2 is constant, i.e., $X_{2i} = X_{2j}$, then $\Delta X_{2i} = 0$ and therefore,

$$\Delta Y_i = \beta_1 \Delta X_{1i}$$

In general, holding all other regressors constant, $\frac{\Delta Y_i}{\Delta X_{1i}} = \beta_1$.

First Differentiate with Respect to the Intercept

Define the residual and the sum of squared residuals:

$$r_i(\boldsymbol{b}) = Y_i - b_0 - b_1 X_{1i} - \cdots - b_k X_{ki}, \qquad Q(\boldsymbol{b}) = \sum_{i=1}^{n} r_i(\boldsymbol{b})^2.$$

The intercept multiplies 1 for every observation, so

$$\frac{\partial r_i}{\partial b_0} = -1.$$

By the chain rule,

$$\begin{aligned} \frac{\partial Q}{\partial b_0} &= \sum_{i=1}^{n} 2 r_i(\boldsymbol{b})(-1) \\ &= -2 \sum_{i=1}^{n} (Y_i - b_0 - b_1 X_{1i} - \cdots - b_k X_{ki}). \end{aligned}$$

At OLS, setting this derivative to zero gives $\sum_{i=1}^{n} \hat{u}_i = 0$.

The OLS Estimator in Multiple Regression

As with simple regression, OLS estimators $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, ..., \hat{\beta}_k$ solve

$$\min_{b_0, b_1, b_2, ..., b_k} \sum_{i=1}^{n} (Y_i - b_0 - b_1 X_{1i} - b_2 X_{2i} - ... - b_k X_{ki})^2$$

The derivative of the sum of squared prediction mistakes with respect to the jth regression coefficient, b_j, is

$$\frac{\partial}{\partial b_j} \sum_{i=1}^{n} (Y_i - b_0 - b_1 X_{1i} - b_2 X_{2i} - ... - b_k X_{ki})^2$$
$$= -2 \sum_{i=1}^{n} X_{ji} (Y_i - b_0 - b_1 X_{1i} - b_2 X_{2i} - ... - b_k X_{ki})$$

Here $j = 1, \ldots, k$; the intercept condition was derived on the previous slide. Combined, these yield the system of $k+1$ equations that, when set to 0, constitute the first-order conditions for the OLS estimator, $\hat{\boldsymbol{\beta}}$ — the $(k+1) \times 1$ dimensional vector of the $k+1$ estimates of the unknown regression coefficients

Two Regressors ($k = 2$): The Three Conditions

Keep n observations, two regressors X_{1i}, X_{2i}, and an intercept. Define

$$r_i(\boldsymbol{b}) = Y_i - b_0 - b_1 X_{1i} - b_2 X_{2i}, \qquad i = 1, \ldots, n.$$

The three first-order conditions are

$$\frac{\partial Q}{\partial b_0} = -2 \sum_{i=1}^{n} (Y_i - b_0 - b_1 X_{1i} - b_2 X_{2i}) = 0,$$

$$\frac{\partial Q}{\partial b_1} = -2 \sum_{i=1}^{n} X_{1i} (Y_i - b_0 - b_1 X_{1i} - b_2 X_{2i}) = 0,$$

$$\frac{\partial Q}{\partial b_2} = -2 \sum_{i=1}^{n} X_{2i} (Y_i - b_0 - b_1 X_{1i} - b_2 X_{2i}) = 0.$$

Evaluate at OLS, $\boldsymbol{b} = \hat{\boldsymbol{\beta}}$, so $r_i(\hat{\boldsymbol{\beta}}) = \hat{u}_i$. At the optimum, the conditions are equivalently

$$\sum_{i=1}^{n} r_i = 0, \qquad \sum_{i=1}^{n} X_{1i} r_i = 0, \qquad \sum_{i=1}^{n} X_{2i} r_i = 0.$$

The Same Three Conditions in Matrix Form

For n observations and $k = 2$ regressors, define

$$\boldsymbol{X}_{n\times 3} = \begin{bmatrix} 1 & X_{11} & X_{21} \\ 1 & X_{12} & X_{22} \\ \vdots & \vdots & \vdots \\ 1 & X_{1n} & X_{2n} \end{bmatrix}, \quad \boldsymbol{b}_{3\times 1} = \begin{bmatrix} b_0 \\ b_1 \\ b_2 \end{bmatrix}, \quad \boldsymbol{r}_{n\times 1} = \boldsymbol{Y}_{n\times 1} - \boldsymbol{X}\boldsymbol{b}.$$

$$\boldsymbol{X}'\boldsymbol{r} = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ X_{11} & X_{12} & \cdots & X_{1n} \\ X_{21} & X_{22} & \cdots & X_{2n} \end{bmatrix} \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_n \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^{n} r_i \\ \sum_{i=1}^{n} X_{1i} r_i \\ \sum_{i=1}^{n} X_{2i} r_i \end{bmatrix}.$$

The prime denotes transpose: $\boldsymbol{X}'$ is $3 \times n$, so $\boldsymbol{X}'\boldsymbol{r}$ is 3×1.

$$\underbrace{\begin{bmatrix} \partial Q/\partial b_0 \\ \partial Q/\partial b_1 \\ \partial Q/\partial b_2 \end{bmatrix}}_{3\times 1} = -2\boldsymbol{X}'(\boldsymbol{Y} - \boldsymbol{X}\boldsymbol{b}); \qquad -2\boldsymbol{X}'(\boldsymbol{Y} - \boldsymbol{X}\boldsymbol{b}) = \boldsymbol{0}_3.$$

Expand the three rows to recover the scalar conditions (next slide).

Each Matrix Row Is One First-Order Condition

For $k = 2$ and n observations, let $\boldsymbol{r} = (r_1, \ldots, r_n)'$, where $r_i = Y_i - b_0 - b_1 X_{1i} - b_2 X_{2i}$.

Row 1: intercept b_0 (slide 14)

$$-2\begin{bmatrix}1 & 1 & \cdots & 1\end{bmatrix}\boldsymbol{r} = -2\sum_{i=1}^{n} r_i$$
$$= -2\sum_{i=1}^{n}(Y_i - b_0 - b_1 X_{1i} - b_2 X_{2i}) = \frac{\partial Q}{\partial b_0}.$$

Row 2: slope b_1 (slide 15)

$$-2\begin{bmatrix}X_{11} & X_{12} & \cdots & X_{1n}\end{bmatrix}\boldsymbol{r} = -2\sum_{i=1}^{n} X_{1i} r_i$$
$$= -2\sum_{i=1}^{n} X_{1i}(Y_i - b_0 - b_1 X_{1i} - b_2 X_{2i}) = \frac{\partial Q}{\partial b_1}.$$

Row 3: slope b_2 (slide 15)

$$-2\begin{bmatrix}X_{21} & X_{22} & \cdots & X_{2n}\end{bmatrix}\boldsymbol{r} = -2\sum_{i=1}^{n} X_{2i} r_i$$
$$= -2\sum_{i=1}^{n} X_{2i}(Y_i - b_0 - b_1 X_{1i} - b_2 X_{2i}) = \frac{\partial Q}{\partial b_2}.$$

At the OLS solution, each derivative equals zero: these are exactly the three entries of $-2\boldsymbol{X}'(\boldsymbol{Y} - \boldsymbol{X}\boldsymbol{b}) = \boldsymbol{0}_3$.

The OLS Estimator in Multiple Regression

For a general sample of n observations, the system of equations can be written in matrix notation as

$$\begin{aligned} -2\boldsymbol{X}'(\boldsymbol{Y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}) &= \mathbf{0}_{k+1} \\ \Rightarrow \boldsymbol{X}'\boldsymbol{Y} - \boldsymbol{X}'\boldsymbol{X}\hat{\boldsymbol{\beta}} &= \mathbf{0}_{k+1} \\ \Rightarrow \hat{\boldsymbol{\beta}} &= (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{Y} \end{aligned}$$

If $\boldsymbol{X}$ has full column rank, $\boldsymbol{X}'\boldsymbol{X}$ is invertible. Where

$$\boldsymbol{Y}_{n\times 1} = \begin{bmatrix} Y_1 \\ Y_2 \\ . \\ . \\ . \\ Y_n \end{bmatrix} \quad ; \quad \boldsymbol{X}_{n\times(k+1)} = \begin{bmatrix} 1 & X_{11} & X_{21} & \dots & X_{k1} \\ 1 & X_{12} & X_{22} & \dots & X_{k2} \\ . & . & . & \dots & . \\ . & . & . & \dots & . \\ . & . & . & \dots & . \\ 1 & X_{1n} & X_{2n} & \dots & X_{kn} \end{bmatrix} \quad ; \text{ and } \quad \hat{\boldsymbol{\beta}}_{(k+1)\times 1} = \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ . \\ . \\ . \\ \hat{\beta}_k \end{bmatrix}$$

Goodness of Fit in Multiple Regression

Remember that:

$$\text{Actual} = \text{Predicted} + \text{Residual} \ (Y_i = \hat{Y}_i + \hat{u}_i)$$

In the previous lectures, we have seen that:

- RMSE = std. deviation of $\hat{u}_i$ *without* d.f. correction
- SER = std. deviation of $\hat{u}_i$ *with* d.f. correction
- R^2 = share of the variance of Y explained by X *without* d.f. correction

Now, we will introduced the adjusted R^2, written $\overline{R}^2$

$\Rightarrow$ $\overline{R}^2$ = share of the variance of Y explained by X *with* d.f. correction

Goodness of Fit: What Changes from $k = 1$?

OLS with an intercept, full rank, and $n > k + 1$; k excludes the intercept.

$$SSR = \sum_{i=1}^{n} \hat{u}_i^2, \qquad TSS_Y = \sum_{i=1}^{n} (Y_i - \bar{Y})^2.$$

Quantity	Simple ($k = 1$)	Multiple (k)
Estimated coefficients	2	$k + 1$
Residual degrees of freedom	$n - 2$	$n - (k + 1)$
RMSE	$\sqrt{SSR/n}$	$\sqrt{SSR/n}$
SER	$\sqrt{\frac{SSR}{n-2}}$	$\sqrt{\frac{SSR}{n-k-1}}$
R^2	$1 - SSR/TSS_Y$	$1 - SSR/TSS_Y$

RMSE and R^2 use the same formulas; residuals and numerical values can change.

- One extra estimated slope uses one extra degree of freedom.
- On the **same observations**, the old fit remains feasible by setting the new coefficient to zero. Thus SSR weakly decreases and R^2 weakly increases; both can remain unchanged.

Adjusted R^2 Also Applies When $k = 1$

Adjusted R^2 adjusts residual variance for the number of estimated coefficients, including the intercept:

$$\overline{R}^2 = 1 - \frac{SSR/(n-k-1)}{TSS_Y/(n-1)} = 1 - \frac{n-1}{n-k-1}(1-R^2).$$

Here $TSS_Y = \sum_{i=1}^{n}(Y_i - \bar{Y})^2$ uses observed Y_i. The $n-1$ counts one estimated Y mean; $n-k-1$ counts $k+1$ estimated coefficients.

$$k = 1: \qquad \overline{R}^2 = 1 - \frac{n-1}{n-2}(1-R^2)$$

The correction already exists in simple regression.

- $\overline{R}^2 \leq R^2$; for fixed k, the difference tends to zero as n grows.
- Adding a regressor can raise or lower $\overline{R}^2$: SSR may fall, but residual degrees of freedom also fall.
- $\overline{R}^2$ can be negative if $SSR/(n-k-1) > TSS_Y/(n-1)$.

For causal analysis, choose controls based on identification, not fit alone.

Reminder: Estimating Coefficient Variances

With an intercept and k regressors, assume full column rank and $n > k+1$.

1. **Estimate error variance from residuals:**

$$\hat{u}_i = Y_i - \hat{Y}_i; \qquad \hat{\sigma}_u^2 = SER^2 = \frac{\sum_{i=1}^{n} \hat{u}_i^2}{n-k-1}.$$

   This estimates the variance of the errors, not the variance of a coefficient.

2. **Replace σ_u^2 in the coefficient covariance matrix:**

$$\mathrm{Var}(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) = \sigma_u^2(\mathbf{X}'\mathbf{X})^{-1} \quad \longrightarrow \quad \hat{V} = SER^2(\mathbf{X}'\mathbf{X})^{-1}.$$

   This $(k+1) \times (k+1)$ matrix has estimated variances of the intercept and slopes on its diagonal.

3. **Read off each variance and take its square root:**

$$\hat{\sigma}^2_{\hat{\beta}_j} = \hat{V}_{j+1,j+1}; \qquad \widehat{SE}(\hat{\beta}_j) = \sqrt{\hat{V}_{j+1,j+1}}, \qquad j = 0, \ldots, k.$$

The Case of $k = 1$

For OLS with an intercept and one regressor ($k = 1$), define

$$TSS_X = \sum_{i=1}^{n} (X_i - \bar{X})^2, \qquad SER^2 = \frac{SSR}{n-2}.$$

$$(\mathbf{X}'\mathbf{X})^{-1} = \begin{pmatrix} \frac{1}{n} + \frac{\bar{X}^2}{TSS_X} & -\frac{\bar{X}}{TSS_X} \\ -\frac{\bar{X}}{TSS_X} & \frac{1}{TSS_X} \end{pmatrix}.$$

The first diagonal entry is for the intercept; the second is for the slope.

Intercept $\hat{\beta}_0$

$$\widehat{\mathrm{Var}}(\hat{\beta}_0) = SER^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{TSS_X} \right), \qquad \widehat{SE}(\hat{\beta}_0) = SER \sqrt{\frac{1}{n} + \frac{\bar{X}^2}{TSS_X}}.$$

Slope $\hat{\beta}_1$

$$\widehat{\mathrm{Var}}(\hat{\beta}_1) = \frac{SER^2}{TSS_X}, \qquad \widehat{SE}(\hat{\beta}_1) = \frac{SER}{\sqrt{TSS_X}}.$$

The same SER^2 multiplies different diagonal entries. Each standard error is the square root of its estimated variance.

The OLS Assumptions for Causal Inference in the Multiple Regression Model

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + ... + \beta_k X_{ki} + u_i \qquad \text{with} \quad i = 1, \ldots, n$$

where $\beta_1, \ldots, \beta_k$ are causal effects and

1. u_i has a conditional mean of 0 given $X_{1i}, X_{2i}, \ldots, X_{ki}$; that is,

$$\mathbb{E}(u_i \mid X_{1i}, X_{2i}, \ldots, X_{ki}) = 0$$

2. $(X_{1i}, X_{2i}, \ldots, X_{ki}, Y_i)$ are i.i.d. draws from their joint distribution.
3. Large outliers are unlikely: $(X_{1i}, X_{2i}, \ldots, X_{ki}, Y_i)$ have nonzero finite fourth moments
4. There is no perfect multicollinearity.

$\Rightarrow$ If assumptions 1-4 hold, then in large samples the OLS estimators $\hat{\beta}_0, \hat{\beta}_1, \ldots, \hat{\beta}_k$ are approximately jointly normal, and each $\hat{\beta}_j \approx \mathcal{N}(\beta_j; \hat{\sigma}^2_{\hat{\beta}_j})$.

What is Multicollinearity?

Multicollinearity refers to the high correlation between two or more independent variables in a regression model.

- **Perfect Multicollinearity:** A linear relationship exists between independent variables (e.g., $X_3 = 0.5 \times$ Share w/ College Education).
- **High Multicollinearity:** High correlation but not a perfect linear relationship (e.g., ρ(Share w/ College Education; Median Income) $= 0.8755$).

Example: dummy variable trap

Suppose you have a set of multiple mutually exclusive and exhaustive variables — that is, there are multiple categories and every observation falls in one and only one category (Freshmen, Sophomores, Juniors, Seniors, Other).

If you include all these variables and a constant (in general, the intercept), you will have perfect multicollinearity — this is sometimes called the dummy variable trap (as it often occurs with dummy variables).

We can then decide to omit one of the groups (or the constant). In this case, the coefficients on the included binary variables represent the incremental effect of being in that category, relative to the base case of the omitted category.

Control Variables in Multivariate Analysis

Definition

A control variable W is a regressor included to hold constant factors that, if neglected, could lead the estimated causal effect of interest to suffer from omitted variable bias.

This definition means that

1. We need to modify the OLS assumptions such that the OLS estimator of the effect of interest is unbiased, but the OLS coefficients on control variables are, in general, biased and do not have a causal interpretation.
2. A good control variable is one which, when included in the regression, makes the error term uncorrelated with the variable of interest.
3. Holding constant the control variable(s), the variable of interest is "as if" randomly assigned.
4. Among individuals (entities) with the same value of the control variable(s), the variable of interest is uncorrelated with the omitted determinants of Y

Conditional Mean Independance

- Because a control variable is correlated with an omitted causal factor, the first OLS Assumption, $\mathbb{E}(u_i \mid X_{1i}, X_{2i}, \ldots, X_{ki}) = 0$ does not hold anymore
- We need a mathematical condition for what makes an effective control variable. This condition is the *conditional mean independence*
- Given the control variable, the mean of u_i does not depend on the variable of interest
- Question: Conditional on sharing the same exposure to adults' educational attainment, is the share of subsidized meals as good as random?

The OLS Assumptions for Causal Inference in the Multiple Regression Model with Control Variables

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + ... + \beta_k X_{ki} + \beta_{k+1} W_{1i} + \beta_{k+2} W_{2i} + ... + \beta_{k+r} W_{ri} + u_i \qquad \text{with} \quad i = 1, \ldots, n$$

where $\beta_1, \ldots, \beta_k$ are causal effects, W_{ji} are control variables, and

1. u_i has a conditional mean that does not depend on the X's given the W's; that is,
$$\mathbb{E}(u_i \mid X_{1i}, X_{2i}, \ldots, X_{ki}, W_{1i}, W_{2i}, \ldots, W_{ri}) = \mathbb{E}(u_i \mid W_{1i}, W_{2i}, \ldots, W_{ri})$$
2. $(X_{1i}, X_{2i}, \ldots, X_{ki}, W_{1i}, W_{2i}, \ldots, W_{ri}, Y_i)$ are i.i.d. draws from their joint distribution.
3. Large outliers are unlikely: $(X_{1i}, X_{2i}, \ldots, X_{ki}, W_{1i}, W_{2i}, \ldots, W_{ri}, Y_i)$ have nonzero finite fourth moments
4. There is no perfect multicollinearity.

$\Rightarrow$ If assumptions 1-4 hold, then in large samples the OLS estimators of interest, $\hat{\beta}_0, \hat{\beta}_1, \ldots, \hat{\beta}_k$ are jointly normally distributed, and each $\hat{\beta}_j \sim \mathcal{N}(\beta_j; \hat{\sigma}^2_{\hat{\beta}_j})$ BUT $\hat{\beta}_{k+1}, \hat{\beta}_{k+2}, \ldots, \hat{\beta}_{k+r}$ might be biased!

Material I

- **Textbooks:**
  - Introduction to Econometrics, 4th Edition, Global Edition, by Stock and Watson – Chapter 6.